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INTRODUCTION TO MACROECONOMICS

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Total number of pages: **12 pages + 1 answer sheet**

Duration of the exam: 120 minutes (30 questions)

Authorized material: none (calculator is not authorized).

Instructions to answer multiple choice questions:

- ★ The answer sheet will be distributed when we proceed with the identity check. Please leave your student card on the table in front of you to facilitate the identity check.
- ★ On this page **AND** on the answer sheet, write down clearly and in capital letters your family name, first name and student number (= 8 digit number without dashes)
- ★ The questionnaire **AND** the answer sheet must absolutely be handed in at the end of the exam.
- ★ On the answer sheet, the box corresponding to the chosen answer (A to D) must be checked or filled properly with a dark blue or black pen.
- ★ If necessary, make corrections carefully. The answer must be clearly identifiable and unique. If this is not the case, it will be counted as wrong.
- ★ We do not answer questions during the exam.
- ★ WC exits are **unauthorized** during the exam.

Example:

Good

	A	B	C	D
Q1	☐	☒	☐	☐
Q2	☐	☐	☒	☐
Q3	☐	☐	☐	☒
Q4	☐	☐	☒	☐
Q5	☐	☒	☐	☐

Bad

	A	B	C	D
Q16	☐	☒	☒	☐
Q17	☐	☒	☐	☐
Q18	☐	☒	☐	☒
Q19	☐	☐	☐	☐
Q20	☒	☒	☐	☐

In order not to waste your chances, answer all questions.
There are no negative points and there is only one correct answer.

Question 1

About the Keynesian and the monetarist view, which of the following statements is true?

- (a) The monetarists believe that only anticipated measures can have an impact on the economy.
- (b) Keynesians believe that the government should have a crucial role during recessions when it should expand its spending.
- (c) One of Keynes' ideas can be simplified as boosting aggregate demand by more government spending in order to end a recession. Nevertheless, he also believed that public debt should be avoided.
- (d) None of the above is correct.

Answer (b)

Question 2

Among the following propositions which one is correct?

- (a) GDP is always measured at factor prices.
- (b) In order to calculate the GDP, capital depreciation has to be taken into account.
- (c) GDP only includes the production of final goods and services.
- (d) GDP of a country does not include goods and services produced by foreign firms inside that country.

Answer (c)

Question 3

Assume that countries A and B can only trade with each other. We dispose of the following information:

Country A		Country B	
Consumption	50	Capital Depreciation	15
Investment	20	Subsidies	10
Government Spending	25	Indirect Taxes	5
Exports	20	National Income	80
Imports	5		
Factor income paid to foreigners	10		
Factor income received from foreigners	15		
Indirect Taxes	10		
Subsidies	unknown		

Additionally, we know that in Country A, capital depreciation equals subsidies minus indirect taxes.

Among the following propositions which one is correct?

- (a) Country A is a net exporter and Country B's GDP at market prices equals 110.
- (b) Country B's GDP at market prices equals 95 and Country A's National Income equals 115.
- (c) Country B is a net exporter and Country A's Net National Product at factor prices equals 110.
- (d) Country B's GDP at market prices equals 95 and Country A's National Income equals 130.

Answer (b)

Question 4

The table here-after gives partial price indices for three groups of goods as well as the weights of the typical bundle (average consumer) for two groups of individuals. Among the following propositions which one is correct?

Group of Goods	Price Index		Weights		
	2010	2011	Typical bundle	Students bundle	Retired workers bundle
Housing	100	120	0.4	0.5	0.2
Food	100	100	0.3	0.3	0.3
Other	100	80	0.3	0.2	0.5

- (a) The retired workers face increasing living standards higher higher than the one observed by the rest of the population.
- (b) The CPI calculated using the typical bundle overestimates the inflation rate for students but is correct for the retired workers.
- (c) The CPI calculated using the typical bundle underestimates the inflation rate for students and retired workers.
- (d) The inflation undergone by students is higher than the one for retired workers

Answer (d)

Question 5

Among the following propositions which one is **false**?

- (a) The Consumer Price Index (CPI) captures the average change in consumer prices from one period to another and the percentage change in the CPI measures the inflation rate.
- (b) The CPI looks at the evolution of prices in the consumption bundle, and the GDP deflator looks at the evolution of prices in the production bundle.
- (c) If the price of imported goods increases more than that of domestic goods, the growth rate of the CPI is higher than that of the GDP deflator.
- (d) The GDP deflator is necessarily increasing from one year to the other.

Answer (d)

Question 6

Which of the following statements is **false**?

- (a) The neo-classical growth model (Solow) theoretically predicts income per capita convergence between rich and poor countries. However, in the data, we cannot observe convergence directly.
- (b) Empirical studies have shown that living standards sooner or later always tend to converge across countries due to increasing returns to capital and decreasing returns to labour.
- (c) The convergence in living standards is empirically observed only when other factors affecting growth, such as education, property rights, legal system, etc., are considered as being relatively similar across countries.
- (d) Very recent empirical studies show that inequality among countries has been declining while inequality within countries has increased.

Answer (b)

Question 7

Which of the following is correct?

- (a) The hypothesis of decreasing marginal returns to factors of production implies that, all other things being equal, countries with lower capital-to-labour ratio grow faster.
- (b) If factors of production exhibit decreasing marginal productivity, doubling the amount of all factors of production cannot double production.
- (c) It follows from the hypothesis of decreasing marginal returns to factors of production that if the endowments of all factors of production double, production doubles as well.
- (d) Decreasing marginal returns to factors of production are not compatible with constant returns to scale.

Answer (a)

Question 8

Consider the following three individuals:

- I. Marie works full time on her PhD thesis in chemistry. She has a scholarship and volunteers in an NGO.
- II. Luc just quitted his full time job as an engineer to spend more time with his kids. He now works from home at 60% for a magazine specialized in home design.
- III. Lise did not call back the director of Human Resources in a fast-food company where she might get a job (he wanted to invite her for an interview).

Which of these three individuals are part of the active labour force?

- (a) All of them.
- (b) Only Luc.
- (c) Only Marie.
- (d) Luc and Lise.

Answer (b)

Question 9

In a closed economy, we know that the GDP is 500, national savings are 50, consumption equals 250 and taxes are 80. Which of the following propositions is correct?

- (a) Investment is lower than private savings.
- (b) Private savings are higher than government expenditure.
- (c) Public savings are equal to private savings.
- (d) None of the above is correct.

Answer (a)

Question 10

In an open economy, national savings can be expressed as

- (a) $S = I$
- (b) $S = Y + G + C$
- (c) $S = I + NX$
- (d) $S = I - C + NX$

Answer (c)

Question 11

Which of the following propositions about the market of loanable funds is correct?

- (a) An increase in the public deficit displaces the demand curve for loanable funds to the left.
- (b) A decrease in the tax on interest revenue results in lower amounts of loanable funds and a lower interest rate.
- (c) A decrease in the public deficit displaces the supply curve of loanable funds to the left.
- (d) An increase in the tax on investment results in lower amounts of loanable funds and a lower interest rate.

Answer (d)

Question 12

Crowding-out describes the effect that

- (a) the government minimizes its market interventions.
- (b) an increase in interest rates due to an increased government deficit reduces private investment.
- (c) a cut in revenue taxes reduces private consumption.
- (d) higher private saving reduces private consumption.

Answer (b)

Question 13

Which of the following is correct?

- (a) Fiat money is the term for money that takes the form of a commodity with intrinsic value such as gold.
- (b) There exist several definitions of money with M0, M1, M2 and M3 the most commonly used. M0 and M1 are less liquid than M2 and M3 because they include saving accounts or time deposits.
- (c) The money supply M1 is affected by the amount deposited in banks and the amount that banks loan.
- (d) The main difference between M1 and M2 is that M1 is made up of tangible money, while M2 is only composed of scriptural money.

Answer (c)

Question 14

The Fisher effect implies that

- (a) fish prices are remaining constant in the long run.
- (b) nominal interest rates increase in the long run.
- (c) if inflation declines, the real interest rate declines.
- (d) if the real interest rate declines, the nominal interest rate declines.

Answer (d)

Question 15

Considering Switzerland as the domestic country, which of the following factors **does not** affect the relation between net exports and real exchange rate?

- (a) A preference shift of the consumers for Swiss chocolate.
- (b) A promotional activity of Valais ski resorts for all residents of the neighboring French and Italian regions.
- (c) The change in the Swiss exchange rate system against the Euro in 2015.
- (d) The interest rate paid on domestic assets.

Answer (d)

Question 16

Which of the following propositions is correct?

- (a) The balance of payments consists only of the balance of trade and the financial account.
- (b) The financial part of the balance of payments accounts for the total stock of foreign assets held by the domestic country.
- (c) The purchase and sale of goods or services are recorded in the current account.
- (d) Exports and imports of securities are recorded in the capital account.

Answer (c)

Question 17

An open economy has a trade deficit. It means that:

- (a) National savings are lower than the investments.
- (b) Net capital outflow is positive.
- (c) The balance of payments balance is negative.
- (d) Private savings are lower than public savings.

Answer (a)

Question 18

Which of the following propositions is correct?

- (a) Under the law of one price, the arbitrage forces equalize the prices of goods sold in the same market.
- (b) The arbitrage forces override the transportation costs.
- (c) The law of one price is not necessarily verified for non-tradable goods.
- (d) The arbitrage forces act on all tradable goods, whether they are perfect substitutes or not.

Answer (c)

Question 19

Suppose the United States exchanges exclusively with Switzerland (domestic country). The money market is initially in equilibrium. Due to a **positive** shock to the credibility of US policy, the capital outflow (from Switzerland) is increasing. Which of the following is correct?

- (a) The dollar demand curve moves to the left.
- (b) The Swiss balance of payments registers a deficit.
- (c) Since there is a financial shock, the American Trade Balance is improving.
- (d) There will be a depreciation of the Swiss Franc against the dollar.

Answer (d)

Question 20

In the context of a floating exchange rate, which of the following is true?

- (a) A deficit in the balance of trade causes a depreciation of the foreign currency.
- (b) A deficit in the balance of trade causes a depreciation of the national currency.
- (c) A positive net outflow of capital causes an appreciation of the national currency.
- (d) The trade balance has no impact on the exchange rate.

Answer (b).

Question 21

Considering a currently unstable economy, what can we say with certainty?

- (a) The capital outflow is increasing, the demand for loanable funds is increasing, the real interest rate is decreasing and the trade balance remains the same.
- (b) The capital outflows is increasing, the demand for loanable funds is increasing, the real interest rate is rising and the trade balance is improving.
- (c) The capital outflow is increasing, the demand for loanable funds is decreasing, the real interest rate is falling and the trade balance is improving.
- (d) The capital outflow is increasing, the demand for loanable funds is decreasing, the real interest rate is falling and the trade balance remains the same.

Answer (b)

Question 22

Which of the following propositions concerning the protectionist trade policies are true?

- I. At the new equilibrium after the introduction of a trade barrier, there is no effect on the balance of trade and the real exchange rate adjusts.
- II. Following the introduction of a restriction on free trade, there is an initial improvement of balance of trade.
- III. Following the adoption of a commercial restriction, there is a temporary decrease in the quantity imported.
- IV. When the country puts into place a restrictive trade policy, the national currency depreciates.

- (a) I, II and III
- (b) I, II and IV
- (c) II, III and IV
- (d) All propositions are correct.

Answer (a)

Question 23

Which of the following facts can explain the link between aggregate demand and the general price level?

- (a) Displayed prices can not be changed instantly (price rigidity).
- (b) Wages adjust gradually (rigidity of wages).
- (c) Changes in domestic prices impact the real exchange rate.
- (d) When the price of a good falls, the demand for that good increases.

Answer (c)

Question 24

Which of the following events will lead to a long-term increase in GDP?

- (a) The marginal propensity to consume decreases.
- (b) The marginal propensity to import decreases.
- (c) The multi-factor productivity decreases.
- (d) The size of the labor force increases.

Answer (d)

Question 25

Suppose that in a country the marginal propensity to consumption is 0.6 and the marginal propensity to import is 0.2. If the government increases its spending by 12 units, the aggregate demand will increase by:

- (a) 12.
- (b) 20.
- (c) 15.
- (d) 60.

Answer (b)

Question 26

Which of the following proposition concerning the impact of economic policies is correct?

- (a) Compared to a flat tax, a proportional taxation of income decreases the Keynesian multiplier.
- (b) In case of a liquidity trap, the fiscal policy is inefficient.
- (c) The Ricardian equivalence argues in favor of an expansionary fiscal policy.
- (d) An expansionary fiscal policy necessarily lowers the interest rate.

Answer (a)

Question 27

Suppose an economy is initially at its long-term equilibrium and it sees its multi-factor productivity increase. If the central bank wants to maintain price stability:

- (a) It must do nothing because multi-factor productivity has no impact on the general level of prices.
- (b) It must do nothing because the adjustment of the anticipated prices will bring the general level of prices back to its initial level.
- (c) It must increase money supply.
- (d) It must decrease money supply.

Answer (c)

Question 28

Suppose an economy is initially at its long-term equilibrium: the unemployment rate is 10% and the inflation rate is 5%. If the central bank decreases the amount of money, which of the following is correct?

- I. In the short term, the unemployment rate is above 10%.
- II. In the long term, the unemployment rate is above 10%.
- III. In the short term, the inflation rate is less than 5%.
- IV. In the long term, the inflation rate is less than 5%.

- (a) Only I and III are correct.
- (b) Only I and IV are correct.
- (c) Only I, III and IV are correct.
- (d) I, II, III and IV are correct.

Answer (c)

Question 29

Consider an economy initially at its long-term equilibrium. Suppose that the expected inflation is 5%, the natural unemployment rate is 10% and (the inverse of) the Phillips curve slope is 0.1 ($a = 0.1$). How much should inflation be to lower the unemployment rate by 1%?

- (a) 15%.
- (b) -15%.
- (c) 5%.
- (d) -5%.

Answer (a)

Question 30

During the 2008 crisis, the monetary stimulus made by the FED was ineffective. Why?

- (a) Banks lent too easily.
- (b) Agents invested all their cash.
- (c) The demand for money was perfectly elastic.
- (d) The unemployment rate was too high.

Answer (c)

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